

LWFA-fed Compton Source Based on Dual-Energy Accumulator Ring with Cryocooled Nb₃Sn Twin-Axis Cavities

Laser plasma acceleration for top-up injection.

Pulsed (fs) laser, pulsed gas jet target. Injection at about 0.2 MeV

Pulsed gas jet for LWFA

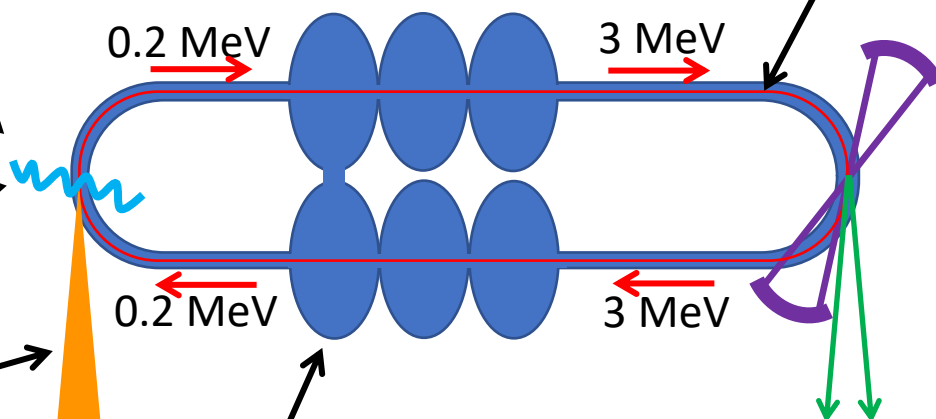
Laser (35 fs, 35 mJ pulse) for LWFA injection

Twin-axis SC cavities, copper covered by Nb₃Sn, working at 4K, cooled by mobile cryocooler. Bottom cells accelerate beam, top cells decelerate beam. Energy gain about 1MeV per cell. Three cells give 3 MeV.

Bunched electron beam

Hypothesis: top-up LWFA injection will allow accumulation of ~Ampere scale current in the ring, helping to boost the source brightness

Laser cavity to store photons (of ~600nm wavelength) for Compton collisions



Compton photons with wavelength ~3 nm (inside “water window”)

Possible staging/building blocks:

- LWFA fs laser injector;
- Pulsed gas jet target;
- Twin-cavities Nb 2K prototype;
- Twin cavities Nb₃Sn at 4K;
- Laser cavity for Compton;
- Diagnostics/Control/Integration;